

CIS111-w02
Computer concepts
Eclipse/run, editor
int variable

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Computer concepts

Chapter 1 – Introduction

Chapter Goals

- To learn about computers and programming
- To compile and run your first Java program
- To recognize compile-time and run-time errors
- To describe an algorithm with pseudocode

Computer Programs

- Computers are programmed to perform many different tasks.
- Computers execute very basic instructions in rapid succession.
- A computer program is a sequence of instructions and decisions.
- Programming is the act of designing and implementing computer programs.
- The physical computer and peripheral devices are collectively called the **hardware**.
- The programs the computer executes are called the **software**.

The Anatomy of a Computer

- Central Processing Unit (CPU) performs
 - Program control
 - Data processing
- Storage
 - Memory (Primary storage)
 - Secondary storage
- Peripherals
 - To interact with human users
- Networks

Figure 1: Central Processing Unit

Figure 2: Hard Disk

Schematic Diagram of a Computer

Figure 3: Schematic Design of a Personal Computer

Computers are Everywhere

- This transit card contains a computer.

The Java Programming Language

- Safe
- Portable
 - Platform-independent
 - Distributed as instructions for a virtual machine
- Vast set of library packages
- Designed for the Internet

Java Versions

- Version 5 (2004): Generic classes, enhanced for loop, auto-boxing, enumerations, ...
- ...
- https://en.wikipedia.org/wiki/Java_version_history

Becoming Familiar with Your Programming Environment

- An editor is a program for entering and modifying text, such as a Java program.
- Java is case sensitive.
- Java compiler translates source code into class files.
- Class files contain instructions for the Java Virtual Machine (JVM).

Becoming Familiar with Your Programming Environment

Figure 6: From Source Code to Running Program

Becoming Familiar with Your Programming Environment

- 1 Start the Java development environment.
- 2 Write a simple program.
- 3 Run the program.
- 4 Organize your work.

Integrated Development Environment (IDE): Eclipse

What is Eclipse?

- Open Source
 - It is a general purpose open platform that facilitates and encourages the development of third party plug-ins
- Best known as an Integrated Development Environment (IDE)
 - Provides tools for coding, building, running and debugging applications
- Originally designed for Java, now supports many other languages
 - Good support for C, C++
 - Python, PHP, Ruby, etc.

Selecting a Workspace

- In Eclipse, all of your code will live under a workspace
- A workspace is nothing more than a location where we will store our source code and where Eclipse will write out our preferences
- Eclipse allows you to have multiple workspaces – each tailored in its own way
- Choose a location where you want to store your files, then click OK

Creating a New Project

- All code in Eclipse needs to live under a project
- To create a project: `File > New > Java Project`

Creating a New Project (continued)

- Enter a name for the project, then click Finish
- Be sure that the box “Create module-info.java file” unchecked

Creating a New Project (continued)

- The newly created project should then appear under the Package Explorer

The src folder

- Eclipse automatically creates a folder to store your source code in called `src`

Creating a Class

- To create a class, simply click on the New button, then select Class

Creating a Class (continued)

- This brings up the new class wizard
- From here you can specify the following. . .
 - Package
 - Class name
 - Whether or not to include a main
 - Etc.
- Fill in necessary information then click Finish to continue

Becoming Familiar with Your Programming Environment

Figure 4: Running the HelloPrinter Program in an Integrated Development Environment

Section_5/HelloPrinter.java

```
public class HelloPrinter {  
    public static void main(String[] args) {  
        // Display a greeting in the console window  
        System.out.println(" Hello , -World!" );  
    }  
}
```

Analyzing Your First Program: Class Declaration

- Classes are the fundamental building blocks of Java programs
- Declaration of a class called `HelloPrinter`:

```
public class HelloPrinter
```

- In Java, every source file can contain, at most one public class.
- The name of the public class must match the name of the file containing the class:
 - Class `HelloPrinter` must be contained in a file named `HelloPrinter.java`

Analyzing Your First Program: Methods

- Each class contains declarations of methods.
- Each method contains a sequence of instructions.
- A method contains a collection of programming instructions that describe how to carry out a particular task.
- A method is called by specifying the method and its arguments.

Analyzing Your First Program: main Method

- Every Java application contains a class with a `main` method
- When the application starts, the instructions in the `main` method are executed
- Declaring a main method:

```
public static void main(String [] args)
{
    . . .
}
```

Analyzing Your First Program: Statements

- The body of the main method contains statements.
- Our method has a single statement:

```
System.out.println(" Hello , -World!" );
```

- It prints a line of text: Hello, World!

Analyzing Your First Program: Method Call

- A method call:

```
System.out.println(" Hello , -World!" );
```

- A method call requires:
 - The method you want to use (in this case, `System.out.println`)
 - Any values the method needs to carry out its task enclosed in parentheses (in this case, `"Hello, World!"`)
 - The technical term for such values is “arguments”

Syntax 1.1 Java Program

Analyzing Your First Program: Strings

- **String**: a sequence of characters enclosed in double quotation marks:

" Hello , - World ! "

Analyzing Your First Program: Printing

- You can print numerical values:

```
System.out.println(3 + 4);
```

- evaluates the expression `3 + 4` and displays the number 7.
- `System.out.println` method prints a string or a number and then starts a new line.
- The sequence of statements:

```
System.out.println(" Hello" );  
System.out.println(" World!" );
```

- Prints two lines: Hello and World!
- There is a second method, `System.out.print`, that you can use to print an item without starting a new line

- A **compile-time error** (syntax error)
 - is a violation of the programming language rules detected by the compiler.

```
System.out.println(" Hello , -World!" );
```

- A **run-time error** (logic error)
 - causes a program to perform an action that the programmer did not intend.

```
System.out.println(" Hello , -Word!" );
```

Errors

- **Exception** – a type of run-time error
 - Generates an error message from the Java virtual machine
 - This statement:

```
System.out.println(1 / 0)
```

- Generates this run-time error message: "Division by zero"

Algorithm

Problem Solving: Algorithm Design

- **Algorithm:** A sequence of steps that is:
 - unambiguous
 - executable
 - terminating

An Algorithm for Solving an Investment Problem

- The problem:
 - You put \$10,000 into a bank account that earns 5 percent interest per year. How many years does it take for the account balance to be double the original?
- Calculating by hand

An Algorithm for Solving an Investment Problem – continued

- The steps in the algorithm:
 - Start with a year value of 0, a column for the interest, and a balance of \$10,000.
- Repeat the following steps while the balance is less than \$20,000:
 - Add 1 to the year value.
 - Compute the interest as $\text{balance} \times 0.05$ (i.e., 5 percent interest). Add the interest to the balance.
- Report the final year value as the answer.

Pseudocode

- **Pseudocode:** An informal description of a sequence of steps for solving a problem
- Describe how a value is set or changed:
 - `total cost = purchase price + operating cost`
 - `Multiply the balance value by 1.05.`
 - `Remove the first and last character from the word.`
- Describe decisions and repetitions:
 - `If total cost 1 < total cost 2`
 - `While the balance is less than $20,000`
 - `For each picture in the sequence`
- Use indentation to indicate which statements should be selected or repeated:
 - `For each car`
 - `operating cost = 10 x annual fuel cost`
 - `total cost = purchase price + operating cost`
- Indicate results:
 - `Choose car1.`
 - `Report the final year value as the answer.`

From Algorithm to Programs

Self Check 1.25

- Suppose you have a random sequence of black and white marbles and want to rearrange it so that the black and white marbles are grouped together. Consider this algorithm:
 - Repeat until sorted:
 - Locate the first black marble that is preceded by a white marble, and switch them.
- What does the algorithm do with the sequence? Spell out the steps until the algorithm stops.
- Answer:

Self Check 1.26

- Suppose you have a random sequence of colored marbles. Consider this pseudocode:
 - Repeat until sorted:
 - Locate the first marble that is preceded by a marble of a different color, and switch them.
- Why is this not an algorithm?
- Answer: The sequence doesn't terminate. Consider the input – the first two marbles keep getting switched.

int variable

Variables

- Most computer programs hold temporary values in named storage locations
 - Programmers name them for easy access
- There are many different types (sizes) of storage to hold different things
 - Will see just `int` for now
- You “declare” a variable by telling the compiler:
 - What type (size) of variable you need
 - What name you will use to refer to it

```
// declaration
```

```
int a;
```

```
// assigning an initial value
```

```
a = 5;
```

```
// declaration and assigning initial value
```

```
int b = 7;
```

```
// use the value of a
```

```
b = a;
```

Syntax 2.1 Variable Declaration

- When declaring a variable, you often specify an initial value
- This is also where you tell the compiler the size (type) it will hold

The Assignment Statement

- Use the “assignment statement” (with an '=') to place a new value into a variable

```
int cansPerPack = 6;    // declare & initialize  
cansPerPack = 8;       // assignment
```

- Beware: The = sign is NOT used for comparison:
 - It copies the value on the right side into the variable on the left side
 - You will learn about the comparison operator in the next chapter